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A Experiment Details

Data Licenses. CounterFact is available by the MIT license at https://github.com/kmeng01/rome [21], and ZSRE is available publicly at http://nlp.cs.washington.edu/zeroshot/ [20].

Data Filtering. We filter the CounterFact dataset to a subset of facts that are correctly completed by GPT-J, in order to ensure that there is knowledge to localize in the model for each point. We mark a completion correct when o_{true} appears among the first 36 tokens sampled from the model given the prompt P using greedy decoding. GPT-J achieves a completion accuracy of 32.6% under this scheme, and after starting with about 10% of the CounterFact dataset, our final sample size is $n = 652$. We perform additional filtering specifically for model editing in the Fact Erasure condition, where we filter points to have a target probability $p_\theta(o_{true}|s, r)$ of at least .02, so that there is a reasonable amount of probability mass to be erased. In this condition, we have $n = 489$ points.

Compute. Experiments were run on a single NVIDIA A6000 GPU with 48gb memory. Computing editing performance for $n = 652$ points with GPT-J for a single edit method applied across model layers in the set {1, 5, 9, 13, 17, 21, 24, 28} could take about eight hours. Saving causal tracing or representation zeroing results for these datapoints takes about twelve hours. Regression analyses and plots can be made on demand (code in supplement) given the data from the editing and localization experiments.

Edit Method Tuning. We tune the edit methods to have high rewrite scores while not trading off too aggressively against paraphrase and neighborhood scores. More specifically, this means we tune methods to have rewrite scores no higher than 99% (note methods can easily get above 99% rewrite score), separately for each editing problem variant. The tuning is done with the first 100 points of the CounterFact dataset, editing layer 6 for GPT-J and 18 for GPT2-XL. For ROME and MEMIT methods, we tune over the KL regularization weight values in the set {.0625, .9, 1}. For constrained finetuning, we tune over the L_∞ norm weight values in the set {1e-4, 5e-5, 2e-5, 1e-5}. For both methods, we adopt default parameters from Meng et al. [22] unless otherwise stated. We describe the relevant hyperparameters below, for GPT-J first:

1. *Error Injection*. FT-1: norm constraint of 1e-4. FT-5: norm constraint of 2e-5. ROME: regularization weight of 1. MEMIT: regularization weight of 0.9.
2. *Tracing Reversal*. FT-1: Norm constraint of 1e-5. FT-5: Norm constraint of 2e-5. FT-5: 2e-5. ROME: default parameters. MEMIT: default parameters.
3. *Fact Erasure*. FT-1: norm constraint of 1e-5. FT-5: norm constraint of 1e-5. ROME: default parameters. MEMIT: default parameters.
4. *Fact Amplification*. FT-1: norm constraint of 1e-5. FT-5: norm constraint of 1e-5. ROME: default parameters. MEMIT: default parameters.
5. *Fact Forcing*. Note that for all methods we decide to increase the number of gradient steps, as convergence takes longer for finetuning (from 25 to 50 steps) and for the gradient-based optimization for v^* in ROME (from 20 to 25 steps). FT-1: norm constraint of 1e-4. FT-5: norm constraint of 1e-4. ROME: 25 gradient steps for finding v^*. MEMIT: default parameters (already set to 25 steps).

We run only the Error Injection and Fact Forcing conditions for GPT2-XL. Hyperparameters are as follows:

1. *Error Injection*. FT-1: norm constraint of 1e-3. FT-5: norm constraint of 1e-4. ROME: default parameters. MEMIT: default parameters.
2. *Fact Forcing*. FT-1: norm constraint of 5e-4. FT-5: norm constraint of 5e-5. ROME: default parameters. MEMIT: default parameters.

B Additional Results

ZSRE Dataset. Here, we describe experiments with the ZSRE dataset, which is commonly used in past editing method papers [9, 24]. ZSRE includes naturalistic questions rather than prompts

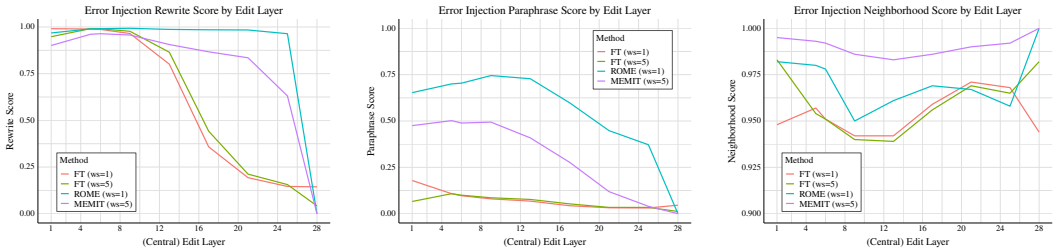


Figure 7: Edit success metrics for our four editing methods, under the Error Injection objective. Left: Rewrite, Center: Paraphrase, Right: Neighborhood.

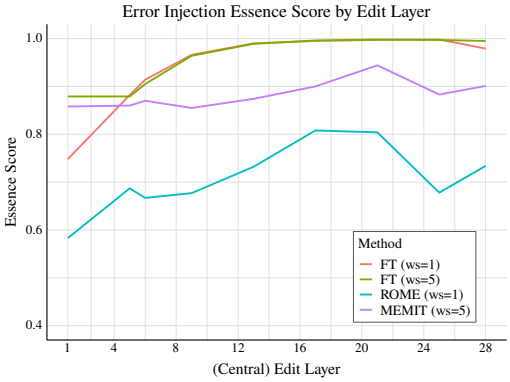


Figure 8: Essence score by edit layer, for our four editing methods, under the Error Injection objective.

intended for autoregressive cloze completion, as in CounterFact. Following past work [21], we use GPT-J to answer ZSRE questions in a zero-shot manner, and we edit the model with ROME. We report results for ZSRE via plots of edit success vs. tracing effect in Figs. 19 (rewrite score) and 20 (overall score), accompanied by regression analysis results in Table 8. We find that results with ZSRE match our conclusions with CounterFact, as the results are quite similar to plots and regressions with CounterFact data. Tracing effects are not predictive of edit success.

Representation Zeroing. Representation zeroing is a common localization technique where neural activations are manually set to zero during a model forward pass [18, 2]. We implement a form of representation zeroing that is exactly like Causal Tracing, except instead of denoising already-noised representations, we set clean representations to zero. Specifically, we simply run a normal forward

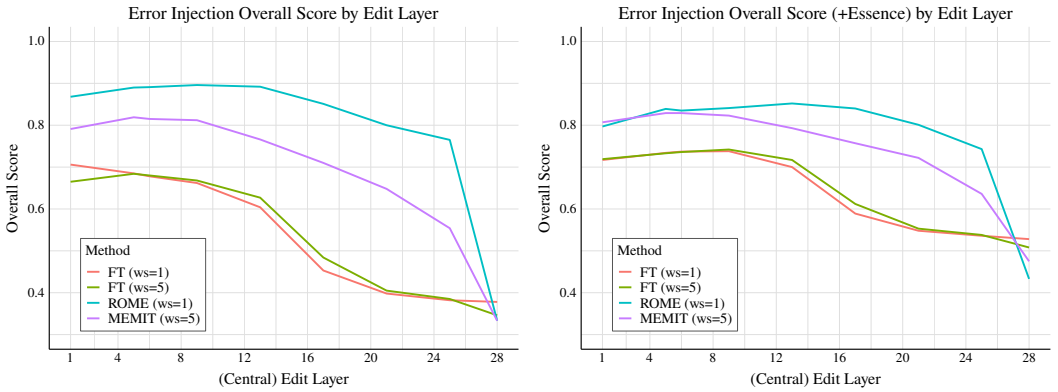


Figure 9: Overall edit success for our four editing methods, under the Error Injection objective. Left: The mean of Rewrite, Paraphrase, and Neighborhood Scores. Right: the mean score with Essence Score included.